CS 210: Worksheet 16 *Dr. Mudassir Shabbir, October 20, 2026*

Student ID 1: _____ **Seat Location 1:** row:_____ col:_____

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Modular arithmetic and equivalence relations. Show reasoning. For $m > 0$, use $a = mq + r$, $0 \le r < m$. To prove equivalence, show reflexivity, symmetry, and transitivity.

1. Quotients and remainders *Easy*

For each (a, m), find $q, r \in \mathbb{Z}$ with $a = mq + r$ and $0 \le r < m$; verify by substitution.

(a) $(a, m) = (-111, 11)$ and $(789, 23)$.

(b) $(a, m) = (-1, 3)$ and $(4, 1)$.

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2. Clock arithmetic *Easy*

On a 24-hour clock (0–23), find each time.

(a) 100 h after 02:00: _____ (b) 45 h before 12:00: _____ (c) 168 h after 19:00: _____

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3. Calculate with congruences *Easy*

Let $a \equiv 4 \pmod{13}$ and $b \equiv 9 \pmod{13}$. Reduce to $\{0, 1, \ldots, 12\}$.

(a) $9a =$_____ $11b =$_____ (b) $a + b =$_____

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4. Find every representative in an interval *Easy*

Let $x \equiv -1 \pmod{25}$.

(a) List all such x with $-100 \le x \le 100$.

(b) Parametrize the class using $k \in \mathbb{Z}$; justify why part (a) is complete.

5. Can you cancel a factor? *Moderate*

The implication $ac \equiv bc \pmod{m} \Rightarrow a \equiv b \pmod{m}$ can fail.

(a) With $m = 6$ and $c = 2$, choose a counterexample a, b; verify the premise and failed conclusion.

(b) From divisibility, prove $2a \equiv 2b \pmod{6} \Rightarrow a \equiv b \pmod{3}$.

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6. Squares rule out a representation *Challenge*

Using remainders, prove:

(a) Every $n \in \mathbb{Z}$ has $n^2 \equiv 0$ or $1 \pmod{4}$.

(b) No $4k + 3$, $k \geq 0$, equals $x^2 + y^2$ for $x, y \in \mathbb{Z}$.

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7. Read equivalence classes from pairs *Easy*

On $A = \{0, 1, 2, 3\}$, let

$$R_1 = \{(0,0), (1,1), (2,2), (3,3)\},$$
$$R_2 = \{(0,0), (1,1), (1,2), (2,1), (2,2), (3,3)\}.$$

$[x] = \{y \in A : (x, y) \in R\}$.

(a) List the distinct classes for R_1 and R_2.

(b) For R_2, find [1] and [2]. Are they equal? Is $0R_23$? Justify from the pairs.

8. Congruence is an equivalence relation *Moderate*

For fixed $m \geq 2$, define aRb on $\mathbb{Z}$ iff $m \mid (a - b)$.

(a) Prove R is reflexive, symmetric, and transitive.

(b) For $m = 6$, give all classes in set-builder form; locate -8 and 10.

9. Recognize a partition *Easy*

For $A = \{1, 2, 3, 4, 5, 6\}$, consider

$$\mathcal{P}_1 = \{\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}\},$$
$$\mathcal{P}_2 = \{\{1\}, \{2, 3, 6\}, \{4\}, \{5\}\},$$
$$\mathcal{P}_3 = \{\{1, 4, 5\}, \{2, 6\}\}.$$

(a) Which $\mathcal{P}_i$ partition A? For each failure, name an overlap or missing element.

(b) For $\mathcal{P}_2$, let $x \sim y$ mean "same block." Find [3] and [4]; decide $2 \sim 6$ and $2 \sim 4$.

10. Bit strings with equal numbers of ones *Moderate*

For finite bit strings, let $s \sim t$ iff they contain the same number of 1s.

(a) Prove $\sim$ is an equivalence relation.

(b) Describe [011]; list its length-4 members. Is the full class finite or infinite? Justify.

11. Equivalent pairs and fractions *Challenge*

For positive integer pairs, define $(a, b) \sim (c, d)$ iff $ad = bc$.

(a) Prove equivalence; in transitivity, state where positivity is used.

(b) Describe $[(1, 2)]$ and interpret all classes as positive rationals.

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12. Compare congruence partitions *Challenge*

Let $[a]_m = \{x \in \mathbb{Z} : x \equiv a \pmod{m}\}$. Partition $\mathcal{P}$ refines $\mathcal{Q}$ if each $\mathcal{P}$-block lies in a $\mathcal{Q}$-block.

(a) Express $[0]_3, [1]_3, [2]_3$ as unions of two mod-6 classes; conclude that mod 6 refines mod 3.

(b) If $m, n > 1$ and $n \mid m$, prove $[a]_m \subseteq [a]_n$ for all $a \in \mathbb{Z}$.

(c) Does $a \equiv b \pmod{3}$ imply $a \equiv b \pmod{6}$? Give a checked counterexample.